

LAB 5: NOISE AND IMAGE RESTORATION

Introduction

Noise is an unwanted change in image intensity that reduces image quality. Different types of noise can affect an image in different ways. Image restoration uses filters to reduce noise and improve the quality of a degraded image.

5.1 Noise

5.1.1 Gaussian Noise

Objective: To generate Gaussian noise and observe its effect on an image.

Theory: Gaussian noise follows a normal distribution. In the given code, mean = 0 and standard deviation = 30 are used to generate and add Gaussian noise to the image.

5.1.2 Rayleigh Noise

Objective: To generate Rayleigh noise and observe its effect on an image.

Theory: Rayleigh noise is generated using the Rayleigh probability distribution. The given code uses a scale value of 80 and adds the generated noise to the image.

5.1.3 Erlang (Gamma) Noise

Objective: To generate Erlang noise and observe its effect on an image.

Theory: Erlang noise is a form of Gamma-distributed noise. The given code uses shape = 3.0 and scale = 20.0 to generate the noise.

5.1.4 Uniform Noise

Objective: To generate uniform noise and observe its effect on an image.

Theory: Uniform noise has values distributed between a lower and upper limit. The given code generates noise between 80 and 180 and adds it to the image.

5.1.5 Salt-and-Pepper Noise

Objective: To generate salt-and-pepper noise and observe its effect on an image.

Theory: Salt-and-pepper noise creates bright pixels with value 255 and dark pixels with value 0. The given code uses 0.3 probability for both salt and pepper noise.

5.2 Image Restoration

5.2.1 Arithmetic Mean Filter

Objective: To reduce noise by replacing each pixel with the average of its neighboring pixels.

Theory: The arithmetic mean filter calculates the average intensity inside a local window. The given code uses a 3×3 window to smooth the noisy image.

5.2.2 Geometric Mean Filter

Objective: To reduce image noise using the geometric mean of neighboring pixels.

Theory: The geometric mean filter calculates the geometric mean of pixel values in a local window. It provides smoothing while maintaining image details better in some situations.

5.2.3 Harmonic Mean Filter

Objective: To reduce noise using the harmonic mean of neighboring pixels.

Theory: The harmonic mean filter uses the reciprocal of pixel values to calculate the local result. The given implementation applies it using a 3×3 window.

5.2.4 Contra-Harmonic Mean Filter

Objective: To reduce salt-and-pepper noise using a contra-harmonic filter.

Theory: The contra-harmonic mean filter uses a parameter Q to control the type of noise it reduces. In the given code, $Q = -2$ is used to reduce salt noise.

5.2.5 Minimum Filter

Objective: To reduce bright noise from the image using the minimum value in a local window.

Theory: The minimum filter replaces a pixel with the smallest value in its neighborhood. It is useful for reducing bright or salt-type noise.

5.2.6 Maximum Filter

Objective: To reduce dark noise from the image using the maximum value in a local window.

Theory: The maximum filter replaces a pixel with the largest value in its neighborhood. It is useful for reducing dark or pepper-type noise.

5.2.7 Median Filter

Objective: To reduce salt-and-pepper noise while preserving image edges.

Theory: The median filter replaces each pixel with the median value of its neighborhood. It is effective for removing impulse noise while preserving edges.

5.2.8 Midpoint Filter

Objective: To reduce noise using the midpoint between the minimum and maximum values.

Theory: The midpoint filter calculates the average of the minimum and maximum values in a local window. The given code first adds uniform noise between -30 and 30 and then applies to the filter.

5.2.9 Adaptive Median Filter

Objective: To remove noise using a median filter whose window size changes according to the image.

Theory: The adaptive median filter changes its window size depending on local image conditions. The given code starts with a 3×3 window and increases it up to 7×7 when required.

Implementation

Source code and Output are attached to this report at the end.

Conclusion

In this lab, five types of noise were generated, and nine image restoration filters were applied. The filters help reduce noise and improve the quality of noisy images using different neighborhood-based methods.